

NETWORKING FUNDAMENTALS AND CIDR BASICS

❖ What is CIDR? (Classless Inter-Domain Routing)

- CIDR stands for **Classless Inter-Domain Routing**. It's a method for allocating IP addresses and routing Internet Protocol packets that CIDR replaces the old **class-based IP addressing system** (Class A, B, C) by allowing **more flexible IP address allocation**.

❖ What is IP is an address?

- An **IP address** (Internet Protocol address) is a unique identifier assigned to a device connected to a network. It allows devices to **communicate with each other** over the Internet or local networks.
 - IPv4 Address: 32-bit address divided into four 8-bit octets (e.g., 192.168.1.1)
 - Written in dotted decimal format (e.g., 192.168.0.1)
 - Each octet ranges from 0 to 255

• Types of IP Addresses

Type	Description
Public IP	Routable over the internet
Private IP	Used within local networks
Static IP	Permanently assigned to a device
Dynamic IP	Assigned by DHCP and can change
Loopback IP	127.0.0.1 – Refers to the local system
APIPA	169.254.0.0/16 – Automatic Private IP Addressing

❖ What is a Subnet Mask?

- A Subnet Mask determines which **portion of the IP address refers to the network** and which part refers to the host.
- It masks the IP **address to identify** the network boundary.

Example:

- IP Address: 192.168.1.10
- Subnet Mask: 255.255.255.0
- **This means:**
 - First 24 bits (255.255.255) → Network portion
 - Last 8 bits (0) → Host portion

❖ Classes of IP:

- **Class A /8** | 0-126 (127.0.0.0 - Localhost / loopback address) | N.H.H.H | 0-255.0-255.0-255 | 256*256*256 | 1C
- **Class B /16** | 128 – 191 | N.N.H.H | 0-255.0-255 | 256*256 | 65536
- **Class C /24** | 192 – 223 | N.N.N.H | 0-255 | 256
- **Class D** | 224 – 239 | Scitific and experimental use
- **Class E** | 240 – 255 | Future Use

❖ Relationship Between IP, Subnet Mask, and Network

- **IP Address :-** Identifies a **device (host)** on a network.
Example: 192.168.1.10

- **Subnet Mask :-** Separates the IP into **network** and host parts. **Example:** 255.255.255.0
- **Network:-** A group of devices sharing **the same network portion** of the IP address **Example:-**192.168.10.0
- **Broadcast Address :-** A **broadcast address** is a special IP address used to **send data to all devices** on a particular network
Example: 192.168.10.255
- **Usable Host Range:-** The **total number of hosts** in a subnet refers to the **total number of IP addresses available** in that subnet — including both usable and reserved addresses. (network + broadcast excluded)

❖ **CIDR Notation :-**

CIDR stands for ***Classless Inter-Domain Routing***. It is a method of allocating IP addresses and routing Internet Protocol packets. **CIDR Notation** is a compact way of writing an IP address **along with its subnet mask**, using a **slash (/)** followed by a number.

❖ **Table: CIDR to Subnet Mask**

CIDR	Subnet Mask	Host Bits	Total IPs	Usable Hosts
/8	255.0.0.0	24	16,777,216	16,777,214
/16	255.255.0.0	16	65,534	65,534
/24	255.255.255.0	8	256	254
/25	255.255.255.128	7	128	126
/26	255.255.255.192	6	64	62
/27	255.255.255.224	5	32	30
/28	255.255.255.240	4	16	14
/29	255.255.255.248	3	8	6

[GENERAL INTERVIEW QUESTIONS]

Q1. What is the difference between TCP and UDP?

- **TCP (Transmission Control Protocol)**
 - **Connection-oriented:** A connection is established before data is exchanged (like a phone call).
 - **Reliable:** Ensures all data is received and in the correct order.
 - **It is a slower**
 - ✓ **Use cases:**
 - Web browsing (HTTP/HTTPS)
 - Email (SMTP, IMAP, POP)
 - File transfer (FTP)
- **UDP (User Datagram Protocol)**
 - **Connectionless:** Sends data without establishing a connection (like sending a letter).
 - **Unreliable:** No guarantee of delivery, order, or error checking.
 - It is Faster and ideal for time-sensitive transmissions.
 - ✓ **Use cases:**
 - Video streaming
 - Online gaming
 - Voice over IP (VoIP)
 - DNS queries

Q2. Explain the significance of the OSI model in networking.

- **The OSI model helps standardize network communication into 7 layers:-** Physical, Data Link, Network, Transport, Session, Presentation, and Application.
- Enables different vendors and systems to **interoperate**

Q3. What is Subnetting?

- **Subnetting** is the process of dividing a large IP network into smaller, more manageable **sub- networks**, or **subnets**.

Q4. Explain the difference between IPv4 and IPv6.

- **IPv4** is 32-bit, supports 4.3 billion addresses, and uses dot-decimal notation (e.g., 192.168.1.1).
- **IPv6** is 128-bit, supports a vast number of addresses, and uses hexadecimal notation (e.g., 2001:db8::ff00:42:8329).

Q5. What is the difference between a public and a private IP address?

- **Private IP** is used within local networks (e.g., 192.168.x.x).
- **Public IP** is used to connect to the internet.

Note:- Private IPs are not routable on the internet.

Q6. How many usable IPs are there in a /30 subnet?

- A /30 subnet provides **4 IP addresses total**, but **only 2 are usable** for hosts.
 - 1] 1 network
 - 2] usable hosts 1
 - 3] Usable host 2
 - 4] broadcast

Q7. What Is a Broadcast Address?

- A **broadcast address** is a allow communication to **all devices in** a subnet.

Q8. What's the difference between a switch and a router?

- A **switch** connects **multiple devices** in a local network.
- A **router** connects **different networks** and manages traffic between them.